

Individual Assignment 3

Choosing the Appropriate Crystal Family Method for Different Driverless Car Features

Introduction

Crystal is not a single process. It is a family of methods that should be tailored to a project's size, criticality, and communication needs. The course material emphasizes that one process will not work for all projects, and that Crystal scales from Clear for small teams to Orange, Red, and beyond as team size and risk increase. It also stresses that Crystal values people, communication, frequent delivery, reflective improvement, and access to expert users over heavy process for its own sake.

For a driverless car product, the right Crystal method must be selected feature by feature rather than for the vehicle as a whole. That is because self-parking, automatic lane change, and honking the horn do not carry the same technical complexity, safety exposure, or staffing need. My overall recommendation is: Crystal Yellow for self-parking, Crystal Orange for automatic lane change, and Crystal Clear for honking the horn when appropriate.

Crystal Family Criteria Used in This Analysis

Three factors for the selection process.

First is **team size**

Second is **criticality**

Third is **process weight**

Summary Recommendation

Feature	Complexity	Risk/Criticality	Likely Team Size	Recommended Method
Honking the horn when appropriate	Low to moderate	Primarily comfort/discretionary/localized mistakes	3–6	Crystal Clear
Self-parking	High	Essential money and safety relevant but usually low speed and bounded	8–15	Crystal Yellow
Automatic lane change	Very high	Safety critical in real traffic and highest impact if wrong	20–30	Crystal Orange

1. Honking the Horn When Appropriate

Recommended Crystal method: Crystal Clear

Complexity: **Low to moderate.** The logic is event driven and narrow in scope compared with full motion planning features

Risk/Criticality: **Usually the lowest of the three.** A false positive horn may annoy or confuse road users while on the other hand a missed horn may reduce warning effectiveness but the feature rarely controls vehicle trajectory directly

Appropriate team size: **3–6 engineers** with one collocated team, plus quick access to domain experts in driving policy, HMI and safety review.

Why this feature matters

This feature sounds simple but it still requires some judgment. The vehicle must decide when a horn is truly appropriate for eg. warning a distracted pedestrian, notifying another driver at low speed or avoiding unnecessary aggressive behaviour. It does not usually require full path generation, complex multi actor negotiation or deep control integration at the same level as lane change.

Why this Crystal method fits

Crystal Clear fits because it is the lightest Crystal method and is intended for small and collocated teams. That matches a compact feature team that can iterate quickly on event classification and user acceptance. The team can rely on osmotic communication and direct feedback from expert users. This is exactly the kind of feature where too much process would slow learning more than it would reduce risk

2. Self-Parking

Recommended Crystal method: Crystal Yellow

Complexity: **High.** The feature combines perception, localization, path planning, obstacle detection, low speed control and environment specific edge cases such as curbs, tight spaces, moving pedestrians and poor markings.

Risk/Criticality: **High** but typically less severe than automatic lane change because operations happen at low speed in constrained environments. Failures may still damage property or undermine trust.

Appropriate team size: **About 8–15 people** across sensing, controls, testing, simulation, and UX/HMI.

Why this feature matters

Self-parking is a serious autonomous capability. The car must find a space, understand boundaries, identify static and moving obstacles, calculate a parking trajectory, and execute steering, braking, and acceleration smoothly. The main reason this feature is not placed above automatic lane change is operational context as it normally occurs at low speed in a bounded parking environment. That lowers exposure compared with high speed traffic but it does not make the feature trivial.

Why this Crystal method fits

Crystal Yellow is the best middle ground. The feature is too complex for Crystal Clear because it needs more than one or two specialists and more coordination across disciplines. Yellow supports a moderate level of process, more structure than clear and a team size that matches the likely staffing for parking perception and test validation.

3. Automatic Lane Change

Recommended Crystal method: Crystal Orange

Complexity: Very high. The feature requires perception of surrounding vehicles, prediction of other actors, gap assessment, motion planning, control execution, compliance with road rules and safe recovery when conditions change during the maneuver.

Risk/Criticality: The highest of the three. A defect here can create a direct safety hazard at medium or high speed. If the system misjudges a vehicle in the blind spot or closes a gap incorrectly, the result can be a collision.

Appropriate team size: Roughly 20–30 people organized across subteams such as perception, prediction, planning, controls, HMI, safety engineering, simulation and external testing.

Why this feature matters

Automatic lane change is the hardest choice in the set and the one that most clearly pushes Crystal upward. It depends on dynamic understanding of vehicles approaching from front and rear, speed differentials, blind-spots condition, road geometry and legal constraints. It also has to coordinate intent signaling and fallback behavior if another actor behaves unexpectedly. This is not just a coding problem rather it is a systems problem.

Why this Crystal method fits

Crystal Orange fits because the lecture material describes Orange for roughly 21–40 people and for projects that need more defined roles and coordination among teams. Orange's stronger structure supports requirements documents, release plans, status reporting, dedicated testing roles, architecture work and multiple sub teams working in parallel

Important caveat: In the lecture we go explicitly go to know that critical projects should not use Crystal methods. In the real world, a fully autonomous highway lane change capability would likely cross into that territory and need stronger safety governance than Crystal alone provides. Therefore, Crystal Orange is the most appropriate choice within the Crystal family because it is the heaviest method discussed in the course material for a feature of this size and risk.

Conclusion

The most appropriate Crystal methods are not the same for all driverless car features. Honking the horn when appropriate should use Crystal Clear because it is the smallest and least complex effort. Self-parking should use Crystal Yellow because it is a high complexity but still bounded feature that fits a medium team. Automatic lane change should use Crystal Orange as it is the most technically demanding and risky feature which requires multiple coordinated subteams and heavier project structure. This feature by feature selection demonstrates the central idea of Crystal i.e. adapt the method to the project's size, criticality and communication needs.